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TRANSCRIPT

Event

“The Nuclear Option: Deciphering Russia’s New Space Threat”

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FEATURING

Mallory Stewart

Assistant Secretary of State for Arms Control, Deterrence, and Stability

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John J. Hamre: Good afternoon, everybody. Glad you're here on a Friday afternoon. Not that many people will do anything on a Friday afternoon that's serious, so it shows that you're all very serious people. And I thank you for coming.

My name is John Hamre. I'm the president here at CSIS. And I want to say a hearty welcome to all of you.

We're in for a very unique opportunity. Assistant Secretary Stewart is – this doesn't normally happen. When there's something that's kind of a breaking, very serious development, we don't normally have the opportunity to hear from a senior policymaker in the State Department or any place to share with us the thinking of the government about a big development, and this is a big development. And so, we're really very fortunate.

I don't know if it was a week ago, 10 days ago Jake Sullivan, in a press availability, made note that we think Russia has put a nuclear device in orbit. Now, I mean, you know, for 50 years we've had what are called RTGs. These are power sources that are used to provide electricity for satellites. They've been nuclear – there's a nuclear core to those. But this is different, OK? This is – this is an implication there's now a weapon in orbit or could be a weapon in orbit, and that's a very profound thing. First of all, it's a violation of international obligations. I mean, that's – but nuclear weapons in space are really profoundly different from nuclear weapons here on, you know, planet Earth. I mean, we're used to thinking about blast effects and there's the shockwave and all that sort of thing. All of that is the product of an earthly environment. But in space, a nuclear detonation has no environment, you know, to transform the energy, and so it becomes X-rays that have astounding reach. And so it's a threat not to something that's five miles away, 10 miles away; it's a threat to things that are a hundred miles away. And when we think of how space is becoming the ubiquitous platform for so many both companies and – countries and companies, this is a profound threat. It's really a serious issue.

And so, we're going to explore that today. And I just want to say thank you to you, Secretary Stewart, for joining us today. It's a – it's a privilege to have you here. I know that many – well, I know government employees are working on a Friday afternoon; not that many of us, you know, civilians are working Friday afternoon.

Clayton is working Friday afternoon. Clayton Swope is going to run this meeting. Let me turn it to you, Clayton. Let's get this started.

Thank you so much for coming, Secretary.

Clayton Swope: Thank you, Dr. Hamre, for that introduction.

Again, my name is Clayton Swope. I am the deputy director of the Aerospace Security Project here at CSIS.

Thank you again, Secretary Stewart, for joining us today – coincidentally, this is National Space Day – to discuss this very important topic. I also want to thank everyone who's here in person and everyone online.

As Dr. Hamre noted, space is an important part of our everyday lives. It plays an increasingly critical role in our economic and national security. As many of those watching may know, the Aerospace Security Project produces an annual report on space threats. We are here today to discuss Russia's development of a nuclear anti-satellite weapon, which was one of the most concerning developments in our most recent report, released just two weeks ago.

So, we've heard a lot of speculation based on few confirmed details about what Russia may be working on since February's news. We know that the United States is working with allies and partners, particularly at the United Nations and through a recent U.N. Security Council resolution, on this issue. So, to start us off, Secretary Stewart, could you tell us more about what the United States assesses that Russia is developing? And how is the U.S. working to address this threat?

The Honorable Mallory Stewart: Thank you, Clayton. And thank you, Dr. Hamre. Really, really happy to be here today on this particular day for May 5th tomorrow, so even more special.

The United States is extremely concerned that Russia may be considering the incorporation of nuclear weapons into its counterspace programs based on information we deem credible. The United States has been aware of Russia's pursuit of this sort of capability dating back years, but only recently have we been able to make a more precise assessment of their progress. Russia has publicly claimed that their satellite is for scientific purposes. However, the orbit is in a region not used by any other spacecraft – that in itself was somewhat unusual – and the orbit is a region of higher radiation than normal lower-Earth orbits, but not high enough of a radiation environment to allow accelerated testing of electronics, as Russia has described the purpose to be.

I also want to take a moment to consolidate here what we have said

previously, just to remind you. What we want to clarify is that this is not an active capability that has already been deployed. Although Russia's pursuit of this capability is deeply troubling, there is no imminent threat. We aren't talking about a weapon that can be used to attack humans or cause structural damage on Earth. Instead, as Assistant Secretary Plumb said yesterday in his testimony, our analysts assess that a detonation in a particular placement in orbit of a – of a magnitude and location would render lower Earth orbit unusable for a certain amount of time.

Of course, the broader issue here is that placement by states parties to the Outer Space Treaty of a nuclear weapon in orbit is a clear violation of the fourth – Article IV of the Outer Space Treaty. And this is something that we've been discussing, we've been emphasizing, and we've been trying to raise awareness of. We also, as Clayton mentioned, pursued a U.N. Security Council resolution to effect – to this effect to try to strengthen the Outer Space Treaty and to try to reinforce states parties' commitment to Article IV, particularly because of the extraordinary result and risk to vital communications, scientific capacity, meteorological capacities, agricultural, commercial, and of course national security capabilities that satellites in this orbit would support and would be immediately eradicated by such a violation and use of such a weapon.

So, you asked what we're doing, and I'd be happy to talk about that. The president has directed a series of actions, specifically with such grave consequences to the – international security and to long-term sustainability of the outer-space environment. We decided to take this issue to the U.N. Security Council. There is no other forum right now for – in the international arena for implementation of the Outer Space Treaty, right? It's different than some of the other arms control treaties in that there isn't a body set up specifically to implement this particular treaty. And given the grave consequences of a potential violation, we thought that there should be a U.N. Security Council resolution reconfirming states parties' commitments to Article IV and taking several other steps, and I can talk about that.

So, one is to consider a U.N. Security Council resolution; at the same time, to engage bilaterally with Russia, bilaterally with other countries. And as Secretary Blinken and I believe Jake Sullivan himself have mentioned, we did discuss this directly with China and with India. And we've been discussing this with other countries that have a vested interest, of course, in the sustainability of the outer-space environment and at the U.N.

So, going to our U.N. proposal, the U.S. and Japan jointly proposed – and

with, ultimately, 65 cosponsors in total – a U.N. Security Council resolution that reconfirmed the importance of the Outer Space Treaty; that recommitted states parties to that treaty to abide by Article IV prohibiting the prevention (sic) in orbit around the Earth of nuclear weapons or other WMD; that reaffirmed the importance of the peaceful uses and sustainable exportation and access to space, and encouraged those member states of the U.N. that weren't states parties to the Outer Space Treaty to consider joining and supporting this; but also, importantly, that called upon states to not develop any nuclear weapons or WMD for placement in outer space. So that went a step beyond, but it was – it was a request calling upon countries to do this. The Outer Space Treaty is unique in that it prohibits the placement in orbit around the Earth of these weapons, but it doesn't talk about the development of the weapons as other arms control treaties do in certain circumstances. So, we thought adding the effort to call upon states to not develop such weapons would be useful.

We thought it was a very nonpolitical, evenhanded approach to remind the world and to reinforce the Outer Space Treaty and its commitments. And we took it forward, again, with a total of 65 cosponsors to the U.N. Security Council. As you know, Russia vetoed it. And you know, as was noted by our ambassador to the U.N., this was inconsistent with Putin's statement that he intended not to place a nuclear weapon in outer space, and we found it raised questions, quite frankly, as to why he felt the need to veto this U.N. Security Council resolution, which also incorporated the idea of additional steps, points that – and I can talk about this in more detail if you want – points that both Russia and China made, the idea of taking additional steps through either legally-binding mechanisms or non-legally-binding to reinforce, to support, and to – and to build upon the commitments in the Outer Space Treaty to prevent an arms race in outer space. Our U.N. Security Council resolution worked to support preventing an arms race in outer space through these mechanisms, especially where we can sort of develop, essentially, credible and reliable means to verify such additional steps.

So, we tried to accommodate the comments we heard from many, both on the U.N. Security Council and off. And we amended the resolution, doubling the number of preambular paragraphs and also adding significant substance to the operative paragraphs, to reflect some of the good comments we heard. We engaged with both Russia and China numerous times. We tried to accommodate and incorporate those suggestions that we could agree to. But they – Russia in particular really relied upon their effort to insist on additional parameters as they proposed in their amendment to the resolution that was not agreed to and as they subsequently proposed in their U.N. Security Council resolution that followed ours that they vetoed.

So, again, next steps. On Monday, May 6th, there's going to be a discussion in the U.N. General Assembly because of Russia's veto to talk about the circumstances for that veto in the U.N. Security Council context, and we really look forward to this as a conversation that brings all members of the General Assembly to hear their thoughts and to understand their concerns on this issue. And then, again, continuing to engage bilaterally and multilaterally, and to use all the diplomatic tools in the contexts that were actually mentioned by both China and Russia, but specifically China in its explanation of vote for our U.N. Security Council resolution the context of the CD, the Conference of Disarmament; the U.N. First Committee; and other diplomatic forums in which these exact issues are being discussed. So, we hope to continue to have these conversations there.

Mr. Swope: Thank you, Secretary Stewart, for that information, what the U.S. is doing and plans to do.

Looking at what language we've heard specifically to date from the U.S. government on this capability, it's been very precise. It's referred to this as a device and not a weapon. It's been couched in very specific terms. Today you just mentioned that one of the red flags with this capability was where it would be placed in orbit. Could you just maybe unpack some of the statements to date on what that really means, and why – what would be the effects on satellites, and why we should really be concerned about it, and how that precise language relates to the U.S. position today?

Hon. Stewart: Yeah. I mean, listen, we always have to be careful when we're talking about releasable language in the context in which, you know, we're dealing with intelligence operations and discussions. And so there has been some carefully chosen language. I ended up reading a little bit more so I can make sure I got the language right, so appreciate you letting me do that.

But you know, what I think John Plumb very effectively communicated yesterday in his testimony is that, you know, the particular placement we're concerned about and our analysts have been looking at would potentially entail, again, the limitation of lower Earth orbit for a large amount or a certain amount of time for all satellites, so the implication for any satellites in that orbital arena if such a weapon were placed with a certain magnitude in a certain location. So, you know, I don't know that we know that much more about the potential consequences; it will depend, again, on all the parameters that our analysts are looking at and what, ultimately, is the placement and the magnitude. But there has been an evolution in our talking points to reflect growing concern on

the issue, but also to reflect the need for more information, right? And I – and I think as we talk with other governments and hear their concerns, and we hear sort of Russian responses, we understand there's a need for as much information as possible so that we're not raising alarm bells without some explanation for why we feel this seriously about this potential threat.

Mr. Swope: And you've mentioned the engagement with other countries, and also at the U.N. but outside of the U.N. What is your biggest takeaway from U.S. efforts to discourage Russia from pursuing this capability? Are there any surprises, particularly with how the vote played out at the U.N.?

Hon. Stewart: Yeah. I mean, thank you for that. We were surprised, actually, to be able to get 65 cosponsors in a fairly short period of time. We were heartened that it was, you know, one country vetoing and one country abstaining at the vote, right, and so 13 countries agreeing with us that our resolution was important for national security, for international stability and security, long-term sustainability of outer space. And I think it reflects the seriousness with which governments are considering this issue that, you know, 13 governments agreed this was the responsible and appropriate approach: again, one country vetoing and one country abstaining. But you know, I think – I think it's really good to know that countries were able to see the potential threat so significantly in such a short period of time.

Mr. Swope: And have you – can you say anything publicly about what China – how China looks at this issue? We haven't heard them say anything or be very vocal about this. It seems like they have just as many space equities –

Hon. Stewart: Yeah.

Mr. Swope: – as the United States in some cases, and as much of a reason economically to want to preserve the use of space. How are they approaching this issue?

Hon. Stewart: Yeah. I'm certainly not going to speak for China, and I should be clear when I say that we got the 13 countries onboard. It's really to reinforce and support the Outer Space Treaty, right, and to recognize the importance of this treaty, because I think, ultimately, the first – you know, the first sort of approach to this is recommitting to an Outer Space Treaty that for almost 60 years has provided the backbone of the sustainable legal architecture for the outer-space environment and for our use and long-term security of that arena.

And so looking specifically to China's response and their explanation of

vote with respect to their abstention, it seems they agreed in the importance of this UNSCR, right, and they specifically said we agree with the need to reinforce and to build upon the structure of the Outer Space Treaty, and they acknowledge and appreciate the, you know, prevention of an arms race in outer space language, the idea in our U.N. Security Council resolution that the CD has also been discussing these issues and it is the one body right now established for legally-binding mechanisms associated with the arms control and nonproliferation architectures here. But they abstained and then supported the Russian amendment. So again, I don't know what their internal thinking is; I can only comment on what they themselves have said. And in their explanation of vote, they said they supported the Russian amendment, which, I should be clear, it specifically calls upon states for all time to prevent the placement of weapons in outer space and then – and then talks about the prevention of force or the threat of force against space, from space to Earth, or Earth to space.

This is a topic that has been actively discussed for many years in the Conference on Disarmament context. We have two open-ended working groups coming up in the next several years to discuss more fulsomely how we prevent an arms race in outer space, but how we really get our heads around are there politically-binding or legally-binding mechanisms to discuss this; how we solve a challenge of definition, which is what is a weapon in outer space – and there is disagreement on that. Could one country use a peaceful satellite for weaponized purposes? How do you – how do you really encapsulate this?

So, China said they wanted to incorporate that amendment. And the challenge there is – several, but they're trying to sort of decide an issue that hasn't been decided even though it's been debated, you know, quite openly and assertively in other contexts for many years. So, several examples. In the CD, we've had in the Open-Ended Working Group on PAROS a conversation about the prevention of placement of weapons in the Outer Space Treaty that Russia and China support. The challenge in that context is it defines weapons in outer space in the – it doesn't – sorry, it doesn't define them; it says we are prohibiting weapons in outer space without a definition of weapon. And the concern for many countries was, what about those Earth-based weapons that can be used in anti-satellite capacities against satellites in space? And in fact, on this front we've seen President Putin himself talk about his space laser in 2018 and 2019, and their deputy foreign minister talk about a subsequent laser that was used both in Ukraine and had the capacity to dazzle satellites 1,500 kilometers in orbit around the Earth. So, they've talked about space-based – I mean, sorry, Earth-based anti-satellite capacities for some time, and so the concern was if you're talking about preventing a placement of weapons in outer space when there's no

definition of weapon, then you're not covering a whole other category of antisatellite capacities. There's a lot of disagreement about how that would play out, what you're trying to do, and how it would be implemented, enforced, or verified, again, in the absence of definitions, in the absence of covering an entire range. But also, the use of force – or threat of use of force in space is a weird thing to limit, given that use of force or threat of use of force is something that the U.N. Charter itself discusses.

And so, there's a lot of debates. And I'm sorry to take us down a deep dive of specifics, but we didn't want the amendment that Russia proposed to decide this issue, which is being actively debated. We specifically limited our resolution to a very clear-cut support for the Outer Space Treaty and the obligations of states parties to that treaty, to limit it from some of the political debates, from ongoing technical debates, and to not decide for countries something that they haven't already agreed upon. And that is what we saw the amendment doing. And that's what China support it in its explanation and vote, and subsequently, of course, the Russian amendment did.

So, China also referenced a previous resolution, 7821, in which they talked about this, you know, prevent the placement of all time? Weapons in outer space – the same exact language. But 50 countries disagreed with that First Committee resolution, and it didn't represent consensus. And there was a significant debate as to what that meant in the context of numerous different weapons and numerous different space and Earth capabilities.

And so, again, trying to limit the Security Council resolution to the specific implementation of the Outer Space Treaty, Article Four, and to remind the world why this was so important, versus taking it down – an issue that hasn't been decided, and should be decided, and should be continued to be discussed, and we encourage that, in the Conference on Disarmament, in the open-ended working groups on PAROS, going forward in the First Committee as those discussions come up, and then the other venues, such as COPUOS, in which these issues are being actively discussed and debated.

Mr. Swope:

Yeah. And I think you raise a really interesting point about the amendment, which is based on the prevention of placement of weapons in space. Which arguably goes beyond what the U.S. and Japan was asking for in the resolution. How does Russia justify that amendment, that position on that particular text, and then not supporting the U.S. and Japanese U.S. Security Council resolution?

Hon. Stewart: Again, you know, I can only speak from their justifications and their explanation of vote. I expect we'll hear much more on May 6th. But their justification essentially is they don't want to agree to reinforce the Outer Space Treaty or Article Four, specifically without incorporating this broader issue of preventing all placement of weapons in outer space. And the challenge for us is that we have presented a resolution that would help strengthen the Outer Space Treaty, remind the world of this commitment, encourage others to join this commitment, and also call on states to prevent the development of these kind of weapons of mass destruction for placement in orbit around the Earth.

So, we think that actually goes as far as we can on issues that collectively can be agreed upon or have been agreed upon. Because, again, it's talking about states parties to the Outer Space Treaty adhering to this and calling upon other countries to take these additional steps. Russia's amendment goes beyond that and wants other countries to agree to things that Russia believes are defined in the appropriate way – again, limiting it to space-based weapons, but not defining what weapon means, and talking about use of force or threat of use of force.

And other countries haven't agreed to that formulation, and nor – nor does that formulation necessarily explain how Russia can continue to threaten the use of land-based, Earth-based weapons against satellites in space. Which, you know, they've done recently. I will highlight your mention of the Space Threat Assessment Report that CSIS does, which is wonderful. And I think it really accurately reflects the challenge and difficulty and complication of all these, you know issues, given other countries' work on space weapons and the potential definitional challenges with respect to these issues.

Mr. Swope: No, that's nice to hear you say. I'm happy to say the name of the report as many times as we can during this, to plug it. (Laughs.) But we're very proud of that work. I'm glad that it means something to you, too.

I want to mention real quickly, we are taking questions from the audience. So, feel free to scan the QR code or to go to our event page and submit them that way. I did want to think about how these fits within the context of Russia's behaviors on arms control in general, maybe disinformation from Russia on issues related to arms control. Do you think that there is a pattern with space-related activities, or with this resolution, that looks similar to how they behave on broader arms control issues?

Hon. Stewart: I mean, I think all of us have been tracking that the Russian Federation has walked away from numerous, if not all, of the existing arms control

and nonproliferation architectures, to a certain degree. Certainly, it's illegal suspension of the New START Treaty, in light of its, you know, ongoing illegal invasion of Ukraine in violation of the U.N. Charter, reflects its desire to place its personal, you know, ambition above international law. And I think we've also seen that in their withdrawal from the CFE. We've seen that in them sort of challenging international architectures, such as the chemical weapons convention in their repeated use of chemical weapons, and in the sort of the broad-based sort of effort to delay and challenge numerous architectures that are existing in the international arena.

In this context what I find most fascinating is that we see in space a reflection of some patterns they've taken elsewhere, which is to, again, threaten, you know, as we saw recently by a Russian governmental official, those space capabilities that support Ukraine, threaten that they may be targeted by the Russian government for supporting Ukraine. So, utilizing threats against space-based capabilities, and also proudly talk about their laser systems, Earth-based ASAT capabilities and, at the same time, put forward the prevention and placement of weapons in our Space Treaty, and this amendment and now resolution reflecting prevention of placement of weapons in outer space, in a way that seems inconsistent. You know, again, preventing threat or use of force – in a manner that seems inconsistent with their activities.

And it's a pattern that we've seen, right? They have utilized chemical weapons and are continuing – as we saw clearly against Navalny and the Skripals – but are continuing to utilize, at least, riot control agents. And, you know, the allegation that was recently made and continued to be investigated about their use of a schedule three chemical weapon in Ukraine, while they purport to support the chemical weapons convention. So, it's – it's a challenge that we've been seeing across the board, actions that are inconsistent with the international legal architecture. So, it seems like a pattern. And I think we should just be careful to try to prevent their effective disinformation narratives from actually continuing to degrade those international architectures that we need to rely upon for our security and the security of the global community.

Mr. Swope: We can move to a couple questions from the audience if that's all right. I have one from Julian Barnes. He's from The New York Times.

How close are the Russians to deploying their anti-space weapon? And what would the diplomatic response to such a deployment be?

Hon. Stewart: I don't have an answer for that, for how close they are to deploying. I think the more important question is how we can work to prevent that.

And I think everything that we're doing in the diplomatic arena is working to prevent the Russians from going forward with this program. And I think the international response should be outrage if this actually does go forward, because it affects everyone, right? Every single country. It's indiscriminate in its potential effect. So, it's much different than efforts to hold us at risk limited to just the U.S. and our partners and allies. It would affect everyone, including China and India, and Russia's own satellites. So, it's extraordinarily destabilizing. And I think the international community needs to reflect that concern.

Mr. Swope: Maybe a follow on to that. If we are looking at ways to stop Russia from this development, does it make sense to look at a capability like this then in terms of an arms control agreement? Is it does it make sense to approach that a different way? With the Security Council work, that's not necessarily a place where a lot of arms control work happens. Is there a new approach that's really needed here because this is so different, or is it look a lot like a typical arms control effort, from the U.S. perspective?

Hon. Stewart: I mean, I think that's a good point to make, is that in the U.N. Security Council resolution context we tried to, you know, reaffirm, and recommit to existing constructs that keep the global community safe. But in our resolution, we proposed, you know, to work towards either politically binding or legally binding mechanisms to further, you know, prevent an arms race in outer space, and work towards some credible way to verify these types of commitments. And I think what we need to figure out in implementing the Outer Space Treaty is how we can develop a capability to credibly establish that Russia is no longer pursuing this program.

And that's a really hard thing to do. It's not something that I think anything – (laughs) – that we can come up with on our own will be likely agreed to by Russia. And I think you've seen in previous contexts where, you know, Syria had denied its chemical weapons program for a long time, until – you know, until basically the U.S. and Russia worked together to bring Syria into the chemical weapons convention and then work to destroy a program that it had denied historically. We're going to have to think of similar creative ideas to figure out how to credibly believe that Russia is not pursuing this program and how to, you know, feel confident that this is not a risk that the world has to continue to focus on.

Mr. Swope: And I have a question here from Neil Wolf at McGill University Institute of Air and Space Law.

And it relates to that international effort to push back on what Russia's

doing. This question is about Japan's participation in the resolution. So, the question is, does Japan's participation in the U.S. resolution reflect Japan's existing or possibly growing role in the ISS, particularly in light of Russia's decreasing ISS involvements? Or is there another reason that Japan would be interested in this topic?

Hon. Stewart: I mean, Japan has been a very close partner with the United States government in space security, but in in the sustainability of outer space. Japan has, I think, deeply supported all of the efforts that the international community has taken to really ensure the long-term sustainability of outer space. And the Japanese space program is clearly doing amazing things to continue to develop and incorporate greater, you know, scientific capacity and improvement of capabilities. And I think Japan was a natural partner and, in fact, really did a really good job in explaining why this should matter to all governments. Not just the U.S. and Japan, but all governments that have present or future reliance on outer space. And I think eventually that will be all governments. And so, Japan was a natural partner for this, and deeply appreciate their pushing this forward.

Mr. Swope: And this is kind of a similar question about the impacts – is there an impact with ISS. But I'll broaden it a bit and talk about the impacts to just human space activities. And this question is from Charles Liu. He's at the City University of New York.

Do you see a direct effect at this time on the International Space Station operations? I'll also add in China's space station, crewed space station, as well as international scientific operations in low Earth orbit or beyond, in space telescope missions or space exploration missions? So, what is the broader – is there a broader impact to these types of activities that are really not national security focused? Definitely focused on that crude aspect, the human aspect? How would – how does this relate to those activities?

Hon. Stewart: Right. So, as I mentioned, this capability is not presently in orbit. However, I think the concern is that it would potentially impact all of the scientific, you know, commercial, communications, agriculture, again, meteorological – all of the capacities in space could potentially be impacted by such a program, if it goes forward to actual placement in violation of the Outer Space Treaty. And, you know, those are impacts that would depend on the magnitude and the location, but that's something that we should all be concerned about.

Mr. Swope: And there's another question along this thread. This is from Joshua Keating from Vox.

How would something like this activity – how would this impact cooperation, what there is left post-Ukraine, between NASA and Roscosmos on ISS or other work? Does it have the potential to really impact what is left of that space cooperation in the civil domain?

Hon. Stewart: Listen, I think it's important that we continue as much cooperation as possible. But I do think, you know, working toward a situation in which we can credibly confirm nothing is going forward that would violate the Outer Space Treaty would be important. But having alerted the international community to this concern, having taken it seriously and addressed it in a U.N. Security Council Resolution, you know, we will continue to push Russia to work with us on this. But, you know, I don't think that the fact that we're working towards credibly confirming that this is not a program that will continue, should necessarily require no cooperation whatsoever. So, as you point out, it's very limited right now. But I don't think that the existing capacity to cooperate, you know, should necessarily be diminished even further, while we try to figure out how Russia credibly demonstrates that it is not pursuing this capability.

Mr. Swope: And we've talked – had a few questions here about the International Space Station, civil efforts in space exploration. One thing I also was wondering your thoughts on, how does a capability like this – how does this maybe cause us to rethink our space architectures? Talking a lot about proliferated constellations and low Earth orbit. Looking at this, is this – is this a way to put those at risk, in your mind? Or is this a different – is this a different goal, if Russia is trying to deploy this capability? And I know, we might not know, but just looking for how the U.S. government looks at why Russia would be doing this, and does this mean that we should rethink how we are looking at our space architectures?

Hon. Stewart: Yeah. I think it's a really good question. There's a lot of things that Russia seems to be doing to potentially have leverage or to hold U.S. government capacities at risk. But in this context, I would definitely refer you to the Department of Defense and to Space Force and Space Command for how we can sort of more effectively respond or address. I think we're doing a lot already to address some of the challenges we have in space, when you talk about sort of these really, you know, diversified responsive architectures. I think they are already efforts that we're taking to address some of the space challenges. But certainly, with respect to how we change or how we evolve, or even what we do to be more responsive and to defend against this, I would refer you to the Department of Defense.

Mr. Swope: Maybe I'll do – we'll do one more question from the audience here, and then start to wrap up. This is from Rameen Skeebah (ph) at Popular Mechanics.

As an effort in transparency, and to nudge Russia to provide more information about this potential weapon, is the U.S. prepared to release more information about its own developing counterspace capabilities?

Hon. Stewart: It's an interesting question because it assumes capacities that, you know, may or may not exist. But certainly, you know, the U.S. has tried to be as transparent as possible with regard to space programs, explaining even in the face of narratives that certain of our capacities are weaponized, when they're not. So, I would refer you to some of the really good communications about what we're doing in space, the clarification and the transparency that's being provided both by the Space Force, and by Space Command, and our Department of Defense colleagues, because I think they have tried to address some of the allegations and trying to demonstrate what actually we're doing in space. So, again, I think there are others that can answer that question better than I can. But I would say, you know, I'm thinking historically, back when some of our extraordinarily NASA-related peaceful scientific efforts were accused of being weapons, you know, we are subjected to allegations all the time. And we try to be as transparent as possible as to the purpose of our programs.

Mr. Swope: Maybe as a question to end on, just looking forward you could see two paths. This is something that Russia develops and deploys this capability, are there concerns that China could want one next? Or this could be a capability that would be proliferating in orbit, more than just Russia wanting this capability? So that's one vision. And I guess the question would be, how could we stop that? You know, is – what are the next steps really that the U.S. is looking for? Is that happening at the U.N.? Is that happening behind the scenes? Is it a combination of that? Realizing that there's probably no silver bullet, how – do you think a scenario like that could happen? And then, you know, what is, in your mind, the best way that the U.S. should be trying to prevent that?

Hon. Stewart: Yeah. I'll use the opportunity to again refer to the Space Threat Assessment Report, talking about normalizing deviance. And that we've seen development in sort of capacities to jam, or to dazzle, or to blind. I mean, the fact that Russia has been bragging about its laser capabilities is normalizing sort of that capacity. We've heard allegations of GPS jamming around Kaliningrad that could potentially impact airlines and the safety of air travel. And so, these kinds of things should not be normalized. But you're right to point out that if we don't prevent them, try to demonstrate them as deviant from what is the responsible

approach, what is consistent with international law, we risk other governments, you know, following – potentially following that path.

We've seen this in numerous arenas in the arms control context, right? Just if we think about chemical weapons, and how serious repeated use of chemical weapons has led to, you know, greater potential use by nonstate actors, and continued use by other governments in ways that demonstrates less and less upholding of the norm. So, it's a context in which arms control often has to operate to redefine and reestablish the importance of international law to prevent, you know, both irresponsible actors, but also illegal activity, and consistent with existing guidelines. So, I think the concern is always that if this is allowed to proceed and become something that gives Russia any sort of advantage, it's harder to prevent other governments from doing the same.

And I think, you know, international law stands, hopefully, as that reminder of something that we've all collectively negotiated because it's in our collective interest. Russia, the U.S., China – everyone's interest to have – to prevent these types of destabilizing capacities. And so that's why we've been working very hard to figure out how we can continue to enforce, and uphold, and support the Outer Space Treaty, and work towards preventing this arms race in outer space by additional mechanisms.

Mr. Swope: Yeah, that's a great way to end this. Just that the Outer Space Treaty, from 1967, a lot of the same motivators back then drive us today. And looking for that safe, stable, sustainable, peaceful space environment is really the same goal. And it sounds like it underpins a lot of your efforts today in how you're approaching this.

Hon. Stewart: Absolutely.

Mr. Swope: Well, we're at time. I want to thank you, Secretary Stewart, for joining us today. Thank you for your support of CSIS generally, particularly your work with the Project on Nuclear Issues led by Heather Williams. Thank you to everyone who joined today in person and online. And with that, have a good weekend, and may the fourth be with you. (Laughter, applause.)

(END.)